ADAGIO GROUP

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Paper 6: Memory Production Review Response

Benjamin D. Summers <bsummers@adagiollc.com> Fri, Sep 18, 2026 at 1:45 AM
To: Ezequiel Boero <ezequiel.boero@adagioinstitute.org>, "ISMAIL-SUTTON, SARA R." <sara.r.ismail-sutton@durham.ac.uk>

Hi Ezequiel and Sara,

Thank you both for your reviews (both attached). The revised manuscript has been uploaded to SSRN (https://dx.doi.org/10.2139/ssrn.7206578). You both reviewed the August deposit without conferring, and you converged on the same points: the scope of the projection-freeness claim, the distinction between the exact deficit-level balance and the charge, the naming of the zero set of production, the two notions of dynamical closure, and the word "memory." My responses follow in manuscript order; where you both raised a point, I answer it once. Section and result numbers are those of the revision; where they differ from the August version I give both. Ezequiel, your Finding numbers below follow your section-by-section analysis, where the numbering differs from your overall assessment.

Ezequiel, thank you for the replication record. I recomputed every figure in your report. Your replication of the paper's displayed numbers, from the coarse kernels and the deficit of Example 3.2 to the joint law, the four norms, and the cross term of Proposition 6.5 and the dial at your five points, agrees with my recomputation in exact arithmetic in every entry. I repeated your two random-family checks, the identities of Lemma 2.3 on twelve rational families with a cut [4] → [2] and the expansion of Proposition 5.1 over all 2^(j−i−1) compositions of the ten pairs of a five-point grid, on my own random rational families in exact arithmetic; every discrepancy is zero. The conditionals of the lumped process of Proposition 6.5 agree to the digit: 37/60, 23/36, 5/8, and the maximal discrepancy 1/24 over the eight conditionings. Your two-minimizer family agrees: the flat families (J/2, I) and (I, J/2) both lie at squared distance 2, the residuals differ by 1/2 in every entry of the first two pairs, and the revision proves that 2 is the minimum and that these are the only minimizers (Example 5.4). Your $h_2(V^{\lambda}) = 0.0265026$ and the ratio 0.9591 reproduce to the printed digit by minimization of the same four-parameter problem. The failure of the conservation identity under source-only weights reproduces as a phenomenon: on 237 random two-step towers the relative residual has median 4.6% and maximum 59%, against 4×10^{-16} in the paper's metric; your range of 4 to 45% differs from mine by sampling variation. Your Finding 3 family and your Finding 4 computation are in the paper.

Proposition 2.6 enters most of the other changes, so I describe it first. Your Question 4 asked whether the constants of the comparison between the charge and the deficit norm (August (7); Proposition 5.3 of the projection paper) can be bounded for n = 2 on a three-point grid. They can, on every grid, with exponent one. For a stochastic K from time a to time b, define the law distortion $\varkappa(K) = \max_v (Kp_a)(v)/p_b(v)$; it lies between 1 and $1/\min p_b$ and equals 1 when K transports the laws. Left multiplication of a kernel difference by K costs at most $\varkappa(K)^{1/2}$ in the law metric, and so does right multiplication (Lemma 2.5). The flat family generated by the one-step kernels of V lies at distance at most $C\|\Delta_V\|$ from V, by telescoping the long kernels against the adjacent deficits, and the deficit of V is at most $L \cdot \mathrm{dist}(V, W)$ for every flat W, by the same two bounds applied to the three terms of $\Delta_V(t,s,r)$. The constants depend only on the grid length m and the smallest law entry ρ: $L^2 = (m-1)(1 + 2/\rho)$ and $C^2 = (m-1) \Sigma_{q=0}^{m-2} \rho^{-q}$. On a three-point grid, C = 1 and $L^2 = 1 + 1/\min p_1 + 1/\min p_2$. So the comparison holds at d = n with exponent one and explicit constants, as (8) and (9) of the revision, and Proposition 5.3 of the projection paper holds at d = n with exponent one; Remark 2.7 records the fact. For Example 3.2 the window is $\sqrt{(54/456125)} \le h_2(V^{\lambda}) \le \sqrt{(18/23575)}$, that is 0.0108 to 0.0277 after outward rounding, and your 0.0265 lies inside it. That answers Question 4 and, with the next paragraph, Question 3.

The charge column (Finding 5, Question 3). I did not add a numerical column. The paper contains no numerical optimization, and I have kept it that way: Table 2 gives the exact window of Proposition 2.6 for every charged family in Table 1, and the paper states that positivity of a charge is certified by exact positivity of the deficit norm. Corollary 4.8 (August 4.7) is restated with the explicit constants, and its lower bound is now $\|\Phi + P\|/L \ge (\|P\| - \|\Phi\|)_+/L$, which is not vacuous in the mixed regime; the corollary is retitled "Charge-level bounds from the balance law" (Sara). The abstract's "the charge inherits the balance law" now reads "the charge inherits two-sided control from the deficit norm with explicit constants and exponent one, so the balance law transfers to the charge as bounds" (Sara's first major comment).

Abstract and significance statement (Sara; Ezequiel Findings 2 and 4, the [R] items on "non-negative" and on the dilation class, Questions 5 and 6). Rewritten. The Mori–Zwanzig sentence carries the flatness hypothesis; the zero set is named two-time Chapman–Kolmogorov lumpability; "dynamically closed" is "closed on the accessible subspace" wherever the zero set is described; "memory" is qualified as the two-time composition defect, and the significance statement ends by saying that the balance law says nothing about the higher-order dependence of the lumped process. The sentence calling production non-negative is deleted; Definition 4.1 says that the production term is a kernel difference, that its magnitude is $\|P\|$, and that it has no sign (Question 5). The dilation class is declared in the abstract,

in Section 1, and in Section 2.1: the time-inhomogeneous class of the projection paper, in which the dimension-matched model set is Flat; in the time-homogeneous class the model set is strictly smaller and (3) fails (Question 6).

Introduction (Finding 6; Sara on the status of the metric). The introduction no longer says that the dependence on the projection does not arise here; it states the three parts of Theorem 5.2 and names the projections the uniqueness statement does not cover. The sentence after (5) says that the balance identity (4) holds in every metric and that the orthogonality in (5), the contraction, and the sizes of the terms are properties of the law-weighted metric, which is where Sara asked for the distinction. A positioning paragraph cites Wolfer and Watanabe, Chruściński, Kossakowski and Rivas, and the Chapman–Kolmogorov test of Markov-state-model validation (Swope, Pitera and Suits; Prinz et al.); the three items are taken up below where they apply.

Section 2.1, the metric (Finding 1, Question 1; Sara on the status of the metric). Accepted, and the sentence was false as printed. The metric (6) is not in the admissible class of the projection paper, whose weights depend on the source state only. It is equivalent to the member of that class with weight p_i(u), with constants (max p_j)^{−1/2} and (min p_j)^{−1/2}, and so to every member; Lemma 4.8 and Proposition 5.3 of that paper need only this equivalence, and the text says so. Proposition 2.6 proves the comparison in (6) itself, so nothing at d = n depends on the companion's class. The paragraph after (6) says which statements are metric-independent (the identities between kernels and the zero sets) and which are properties of (6) (the orthogonality, the contraction of the flux, and the sizes of the terms); Remark 2.4 says that the conditional-expectation structure fails for the companion's class. That answers Question 1: the two are not members of one class; they are equivalent norms, and the paper cites equivalence.

Section 2.2, "compatible with the law" (Sara). Accepted; the term is gone. Definition 2.1 says the laws are strictly positive and the cut is surjective.

Section 2.3, Lemma 2.3 (Sara). Accepted. The preamble names the spaces: $H = L^2([N]^2, p_j \otimes p_i)$, its subspace H_λ of B_λ-measurable functions, and $H^\lambda = L^2([n]^2, p^\lambda_j \otimes p^\lambda_i)$; the density identifications are isometries, the pull-back is an isometry of H^λ onto H_λ, and the compression is the composition $H \to H_\lambda \to H^\lambda$ of the conditional expectation with the inverse pull-back, not an operator on one space. The phrase "metric surjection" is gone; part (c) says that C_i maps H onto H^λ with the contraction and the Pythagorean identity. Remark 4.3 adds the reading of the severed component that the lemma gives: it is orthogonal to the block-measurable subspace for the chosen cut in the metric (6), invisible to observables measurable with respect to that partition and not to every coarse observation (Sara), and $\|\Xi(t,s,r)\|^2$ is the mean conditional variance of the deficit density given the coarse partition (Question 16).

Section 3 (Sara on Example 3.2 and on "metric size"). Accepted. Example 3.2 says that the positive charge follows from the nonzero deficit and not from the failure of strong lumpability, and gives the exact window. The charge is described as a distance-based measure of the Chapman–Kolmogorov failure and $\|\Delta_V\|$ as its metric size, in the introduction and in the significance statement, with Proposition 2.6 as the relation between them.

Section 4, closure (Sara). Accepted. Definition 4.4 defines both notions, dynamical closure on all of $\mathbb{R}^N$ and closure on the accessible subspace, and states that the second follows from the first and that the converse can fail; Theorem 4.5 is unchanged in content, and the prose says "closed on the accessible subspace" wherever it means the zero set. The new Example 4.7 shows the converse failing.

Corollary 4.6 (Ezequiel's [R] item, Question 10; Sara). Accepted with Ezequiel's term. The corollary is "Two-time Chapman–Kolmogorov lumpability," the condition is defined in the statement, and the relation to weak lumpability is stated in the corollary. Weak lumpability in the Kemeny–Snell sense, Markovianity of the lumped process for the given initial law, implies it, strong lumpability implies it, and neither implication reverses. That answers Question 10. Sara, your recommendation was to keep "in its two-time formulation" consistently; I adopted the rename instead, because there is no established two-time formulation of weak lumpability for the qualifier to refer to. In writing the corollary I found that the paper had no witness for the first non-implication under the corollary's own hypothesis, a flat fine family: Feller's example is not a lumping, and the interference example has a charged fine family. Example 4.7 supplies one: Y_0, Y_1 independent fair bits, $Y_2 = Y_0 \oplus Y_1$, lifted to the deterministic four-state chain $X_0 = (Y_0, Y_1)$, $X_1 = (Y_1, Y_0)$, $X_2 = (Y_0 \oplus Y_1, Y_1)$ with the first coordinate as the cut. Every coarse kernel is J/2, so the coarse family composes and production vanishes, while $P(Y_2 = 0 \mid Y_1 = 0, Y_0 = 0) = 1$ against 1/2. The same chain is closed on the accessible subspace at level 1 and not dynamically closed there. Ezequiel, the sentence of the August proof that you singled out, on the gap between pairwise composition and the Markov property, is now the non-implication in the statement of the corollary and the example; it was not dropped.

Section 4, the long kernel (Question 14). Yes. Proposition 4.9 states that production does not depend on $V(t \leftarrow s)$, that the flux is affine in it, and that for fixed short kernels the coherence condition is a closed half-space condition on the compression of the long kernel; the sign question is linear in the long kernel, and the dial of Remark 6.6 is its one-dimensional case. Open Problem (i) is restated in that form.

Section 5.1 (Sara on Proposition 5.1; Ezequiel on backflow, Question 11). Accepted. A paragraph after Proposition 5.1 states the correspondence with the projection-operator formalism term by term (the all-Π word as the resolved evolution, the Q-runs as orthogonal dynamics, the segment kernels as the memory kernels) and states its limits: the objects are conditional kernels and not observables, there is no fluctuating force, and no continuous-time limit is taken. The backflow sentence is an analogy, and the inequivalence is cited: divisibility implies the absence of backflow and the converse fails. Question 11: no.

Section 5.2 (Finding 2, Questions 2 and 9; Sara). Accepted, both narrowings. Theorem 5.2 keeps its hypothesis, and the abstract, the introduction, and the section carry it. The section is retitled "The law section among admissible sections"; the text says that the deficit of the observed family is defined without a projection, that the class over which uniqueness is proved is the block-supported stochastic sections, and that Mori's linear projections and the Wiener projections are not of that form. Question 2: yes, both places carry the hypothesis. Question 9: no, the class does not contain them; the paper says what is proved and no more.

Remark 5.3 and Example 5.4 (Finding 3, Question 8). Accepted, your first repair. Canonicality is claimed for h_n alone. The residual at a nearest flat family has norm h_n and is defined up to the choice of minimizer; the remark says that the metric projection onto Flat_n is not unique in general, that the non-uniqueness set has Lebesgue measure zero on the affine hull, and that the paper uses only h_n. Your family is Example 5.4, with a proof that the minimum is 2: with $x = 1 - a_1$, $y = a_2$, $u = 1 - b_1$, $v = b_2$, the squared distance is $2(x^2 + y^2 + u^2 + v^2 + c_{11}^2 + c_{22}^2)$ with $c_{11} + c_{22} = 1 + (1 - \alpha)(1 - \beta)$ for $\alpha = x + y$ and $\beta = u + v$, and three elementary inequalities reduce the sum to $1 + [(1 - \varsigma)^2 + \tau^2]/2$ with $\varsigma = (1 - \alpha) + (1 - \beta)$ and $\tau = (1 - \alpha)(1 - \beta)$, which is at least 1 with equality only at your two families. The example also shows that the upper bound of Proposition 2.6 can be strict: $\|\Delta_V\| = 2$ and $h_2 = \sqrt{2}$. Question 8: in no sense; the norm is canonical and the vector is not.

Section 6 (Sara on Theorem 6.2, on "grow only through production," on "no autonomous factor," and on the dial; Ezequiel Finding 4, Question 7). Accepted. Theorem 6.2 is "Monotonicity of the deficit norm," its charge chain is written with the explicit constants, and the text says that the chain does not imply monotonicity of the charge and that positivity is preserved. Remark 6.3 says that any increase of the coarse deficit norm beyond the transmitted fine deficit is supplied by the production term and that nonzero production does not by itself increase the norm. Proposition 6.5(i) says that the cut is not an autonomous-factor projection of the type of Proposition 6.4, and no more. Remark 6.6 gives the exact admissible intervals of the dial, $x \in [-5/32, 5/16]$ for the kernel and $[-3/32, 3/16]$ for the joint law, and says that the family is collinear and that the general balance law allows arbitrary orientations. Finding 4 is Remark 6.8: from the displayed joint law, the lumped process has $P(Y_2 = a \mid Y_1 = a, Y_0 = a) = 37/60$ and $P(Y_2 = a \mid Y_1 = a, Y_0 = b) = 23/36$ against $5/8$, with maximal discrepancy $1/24$, so the interference cut cancels the two-time composition defect and leaves the dependence of the lumped process on its past in place; the remark says that every occurrence of "memory" in the paper refers to the two-time defect. That answers Question 7.

Section 7 (Sara on exact verification and on "four pure regimes"; Ezequiel Finding 5). Accepted. The section is "Exact examples." The rows are called illustrations and consistency checks of Theorem 4.2, which is proved in Section 4; the four rows are extremal mechanisms, and the text says that the general cut does not fall into one of the four classes. Table 2 gives the charge windows.

Section 8 (Sara). Added: a conclusion that states the exact object, the three mechanisms, the way the charge inherits control, and the limits of the results, namely the two-time scope, the flatness hypothesis on the Mori–Zwanzig identification, the uniqueness class, the non-uniqueness of the residual, the metric-dependence of the geometric statements, and the restriction to the dimension-matched charge.

Section 9, open problems (August Section 8; Sara on (iii), (iv) and (v); Ezequiel's [R] items on the plug-in claim and on the test, Questions 12, 13, 15 and 16). Accepted. (i) is restated through Proposition 4.9. (ii) states the balance law for the full profile h_d as open and records why the argument of Proposition 2.6 does not extend to $d > n$: the model sets there are closures of compressions and not graphs over one-step kernels (Question 15). (iii) poses the scaling of deficit and production with the time increment as the first problem, calls the relation to entropy production an analogy, and adds the question of what the severed term captures of higher-order dependence. (iv) is a test of two-time closure and not of Markovianity; the plug-in statement holds with the laws fixed, and with estimated laws it needs a margin on the block masses and joint propagation of the law error, since the disintegrations divide by those masses. Ezequiel, I accept that the bounds of Corollary 4.8 are not a test; the text now says that they relate the statistic to the charge, that a test needs the null distribution of the statistic when the kernels and the laws are both estimated, and that this is open. The empirical counterpart is the Chapman–Kolmogorov test of Markov-state-model validation (Question 13). (v) says that minimizing production and minimizing the charge have the same zero set and are two-sidedly comparable, and need not order the cuts identically. Question 12: the same geometric content, an orthogonal split relative to a projection onto coarse structure, in a different geometry; the production term has no counterpart there. Question 16: yes, as the mean conditional variance of the deficit density given the coarse partition; no relation to the divergence-based lumping losses is claimed.

Not adopted. The numerical charge column, for the reason above. Everything else in both reviews is in the revision.

Bibliography: I added Wolfer and Watanabe 2024; Chruściński, Kossakowski and Rivas 2011; Swope, Pitera and Suits 2004; Prinz et al. 2011; and Feller 1959. I updated the companion citations: the foundations paper under its new title, "The Physics of the Instant," and the Koopman, convex-perturbation, selection-functional and projection papers to their September revisions. Every result cited from the projection paper retains its number in that paper's revision 2.

Let me know your thoughts.

Ben

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On Tue, Sep 15, 2026 at 2:57 PM Ezequiel Boero <ezequiel.boero@adagioinstitute.org> wrote:

> Hi Ben,
> attached is the last technical report - paper 6 "Memory Production under Coarse-Graining: A Balance Law for the Indivisibility Deficit"
>
> I'll proceed to integrate the reports, reply to your responses and
> finish the report.
>
> Best,
> Ezequiel

2 attachments

Paper 6- Memory Production EB Review.pdf
264K

Paper 6 - Memory Production SI Review.pdf
156K